Since this is your first traffic light project, here's a complete step-by-step guide from hardware assembly to testing.

Project Objective

Create a traffic light system where:

- 🔴 Red LED is ON by default.
- When a car is detected by the IR sensor:
    - 🔴 Red turns OFF.
    - 🟡 Yellow turns ON for **3 seconds**.
    - 🟢 Green turns ON.
- After a specified time (e.g., 10 seconds), the system returns to:
    - 🔴 Red ON.

Step 1: Collect Hardware

Component	Quantity
Arduino Uno/Nano	1
IR Obstacle Sensor	1
Red LED	1
Yellow LED	1
Green LED	1
220Ω Resistor	3
Breadboard	1
Jumper Wires	As required
USB Cable	1
5V Adapter (Optional)	1

Step 2: Build the Circuit

IR Sensor Connections

IR Sensor	Arduino
VCC	5V
GND	GND
OUT	D2

LED Connections

Red LED

```
Arduino D9
            |
   220Ω
            |
 (+) Red LED
 (-)
            |
     GND
```

Yellow LED

```
Arduino D10
              |
    220Ω
              |
 (+) Yellow LED
 (-)
              |
      GND
```

Green LED

```
Arduino D11
              |
    220Ω
              |
 (+) Green LED
 (-)
              |
      GND
```

Step 3: Verify the Wiring

Before connecting power:

- Check all GND connections.
- Ensure each LED has its own 220 Ω resistor.
- Verify LED polarity:
  - Long leg (Anode) → Arduino pin through resistor.
  - Short leg (Cathode) → GND.
- Verify the IR sensor OUT pin goes to D2.

Step 4: Install Arduino IDE

Download and install the Arduino IDE.

Install the board package if required:

- Arduino Uno: no extra package.
- Arduino Nano: select the correct processor if using a clone.

Step 5: Connect the Arduino

- Connect the USB cable.
- Open Arduino IDE.
- Select:
    - Board → Arduino Uno (or Nano).
    - Correct COM Port.

Step 6: Upload the Program

The program should:

1. Keep Red LED ON.
2. Read the IR sensor continuously.
3. When a vehicle is detected:
    - Turn Red OFF.
    - Turn Yellow ON for 3 seconds.
    - Turn Green ON.
4. Wait for the green duration (e.g., 10 seconds).
5. Turn Green OFF.
6. Turn Yellow ON for 2 seconds.
7. Turn Red ON again.

Step 7: Test the IR Sensor

Open the Serial Monitor.

Move your hand in front of the IR sensor.

The built-in LED on the sensor should change state when it detects an object.

If it doesn't:

- Adjust the blue potentiometer on the IR sensor.
- Keep the object within 5–20 cm.
- Ensure the sensor is receiving 5 V.

Step 8: Test Each LED Individually

Before testing the full program, upload a simple sketch that turns each LED ON one by one.

Example sequence:

```
Red ON
↓
Yellow ON
↓
Green ON
↓
Repeat
```

This confirms all LEDs and wiring are correct.

Step 9: Test Vehicle Detection

Place a toy car or your hand in front of the sensor.

Expected sequence:

```
Normal

□ ON

↓

Car Detected

□ OFF

↓

□ ON (3 sec)

↓

□ ON (10 sec)

↓

□ ON (2 sec)

↓
```

▫ ON

Step 10: Final Demonstration

Arrange the LEDs vertically to resemble a real traffic signal:

-

▫

▫

Mount the IR sensor so it faces the road or the model vehicle.

Use a toy car to demonstrate the automatic signal change.

Working Flow

```
Power ON
      │
      ▼
Red LED ON
      │
      ▼
Read IR Sensor
      │
      ├── No Vehicle
      │       │
      │       ▼
      │   Keep Red ON
      │
      ▼
Vehicle Detected
      │
      ▼
Red OFF
      │
      ▼
Yellow ON (3 sec)
      │
      ▼
Green ON (10 sec)
      │
      ▼
Yellow ON (2 sec)
      │
      ▼
Red ON
      │
      ▼
Repeat
```

Suggested Improvements

Once the basic system works, you can add:

- 🚦 Pedestrian push button
- ⏱ 7-segment countdown timer
- 🖥 OLED display showing "STOP", "READY", "GO"
- 🚗 Two-way intersection with two traffic lights
- 🚨 Emergency vehicle priority using RFID or Bluetooth
- 📡 IoT monitoring using ESP32 and Wi-Fi
- 🚗 Multiple IR sensors to adjust green time based on traffic density

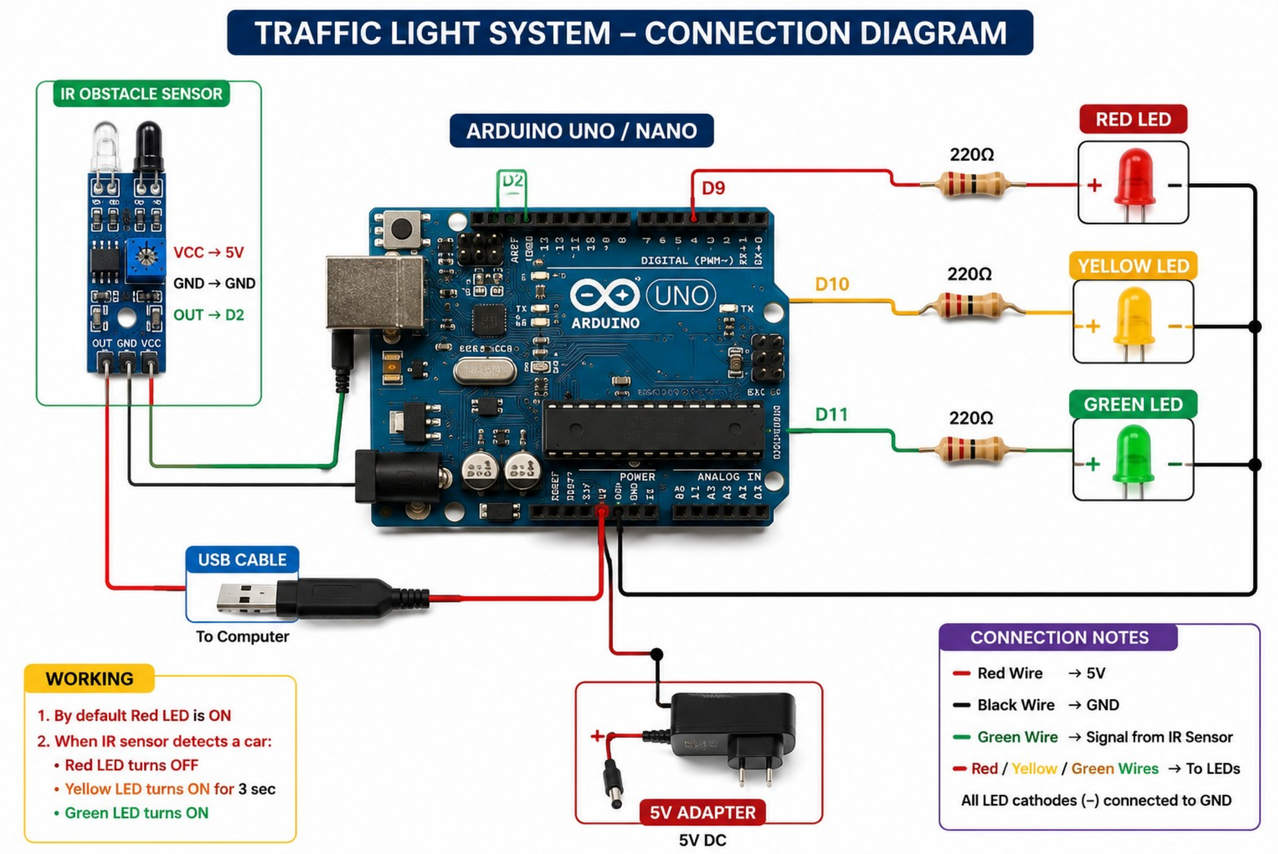
Project Cost (Approximate)

Component	Approx. Price (INR)
Arduino Uno Clone	₹350–450
IR Obstacle Sensor	₹80–120
3 LEDs	₹15
3 × 220 Ω Resistors	₹5
Breadboard	₹100–150
Jumper Wires	₹80–100
USB Cable	₹50–100

Estimated total cost: ₹700–₹900, depending on where you purchase the components.

This setup is reliable, easy to assemble, and provides a solid foundation for adding more advanced traffic management features later.

BLOCK DIAGRAM :


TRAFFIC LIGHT SYSTEM – CONNECTION DIAGRAM
IR OBSTACLE SENSOR
VCC → 5V
GND → GND
OUT → D2
OUT GND VCC
ARDUINO UNO / NANO
D2
D9
D10
D11
DIGITAL (PWM~)
UNO
ARDUINO
POWER
ANALOG IN
220Ω
220Ω
220Ω
RED LED
YELLOW LED
GREEN LED
USB CABLE
To Computer
5V ADAPTER
5V DC
WORKING
1. By default Red LED is ON
2. When IR sensor detects a car:
• Red LED turns OFF
• Yellow LED turns ON for 3 sec
• Green LED turns ON
CONNECTION NOTES
Red Wire → 5V
Black Wire → GND
Green Wire → Signal from IR Sensor
Red / Yellow / Green Wires → To LEDs
All LED cathodes (–) connected to GND